

Characterizing fault-tolerant hamiltonian properties with degree sequences *

Shih-Yan Chen¹, Hsun Su², and Shin-Shin Kao^{3†}

¹Taipei Municipal Bai Ling Senior High School,
Taipei, Taiwan 11167, R.O.C.

²Department of Public Finance and Taxation,
Takming University of Science and Technology,
Taipei, Taiwan 11451, R.O.C.

³Department of Applied Mathematics,
Chung Yuan Christian University,
Chung-Li City, Taiwan 32023, R.O.C.

Abstract

Let n and δ be positive integers with $2 \leq \delta < n$. Given a graph $G = (V, E)$ with $|V| = n$, and $F \subset V$ with $0 \leq |F| \leq k$. G is called k -vertex fault traceable, k -vertex fault hamiltonian, or k -vertex fault hamiltonian-connected if $G - F$ remains traceable, hamiltonian, and hamiltonian-connected, respectively. Let the minimum degree $\delta(G) \geq \delta$. In this paper, we establish a theorem which uses the degree sequence of a given graph to characterize the k -vertex fault traceability/hamiltonicity/hamiltonian-connectivity, respectively. It improves and extends the known results for $k = 0$.

1 Introduction

In this paper, we denote any graph by $G = (V, E)$, where V is the vertex set and $E \subseteq \{(u, v) \mid (u, v) \text{ is an unordered pair of } V\}$ the edge set of G . The order of a graph G , denoted by $|G|$, is the number of vertices of G . An edge of G is denoted by (u, v) , where $u, v \in V$ and $(u, v) \in E$. Two vertices u and v are adjacent in G if there is an edge (u, v) in G . The degree of a vertex u in G , denoted by $\deg_G(u)$, is the number of vertices adjacent to u . The notation $\delta(G)$ represents the minimum degree of vertices of the

graph G . A path P between two vertices v_0 and v_k is represented by $P = \langle v_0, v_1, \dots, v_k \rangle$, where each pair of consecutive vertices are connected by an edge. A hamiltonian path between u and v , where u and v are two distinct vertices of G , is a path joining u to v that visits every vertex of G exactly once. A graph G is traceable if G contains a hamiltonian path. A cycle is a path of at least three vertices such that the first vertex is the same as the last vertex. A hamiltonian cycle of G is a cycle that traverses every vertex of G exactly once. A hamiltonian graph is a graph with a hamiltonian cycle. A graph G is connected if there is a path between any two distinct vertices in G and is hamiltonian-connected if there is a hamiltonian path between any two distinct vertices in G . A graph G is k -vertex fault traceable if $G - F$ remains traceable for any set $F \subseteq V(G)$ with $|F| \leq k$, where $k \leq \delta(G) - 1$. A graph G is k -vertex fault hamiltonian if $G - F$ remains hamiltonian for any set $F \subseteq V(G)$ with $|F| \leq k$, where $k \leq \delta(G) - 2$. A graph G is k -vertex fault hamiltonian-connected if $G - F$ is hamiltonian-connected for any set $F \subseteq V(G)$ with $|F| \leq k$, where $k \leq \delta(G) - 3$ [8].

Let G_1 and G_2 be two graphs. We say that G_1 and G_2 are disjoint if G_1 and G_2 have no vertex in common. The union of two disjoint graphs G_1 and G_2 , denoted by $G_1 + G_2$, is a graph with $V(G_1 + G_2) = V(G_1) \cup V(G_2)$ and $E(G_1 + G_2) = E(G_1) \cup E(G_2)$. The join of two disjoint subgraphs G_1 and G_2 , denoted by $G_1 \vee G_2$, is the graph obtained from $G_1 + G_2$ by joining each vertex of G_1 to each vertex of G_2 .

*This research was partially supported by the National Science Council of the Republic of China under contract NSC: NSC 102-2115-M-033 -004 -.

†Correspondence to: Professor S.-S. Kao, e-mail: shin2kao@gmail.com.

We use K_n for a complete graph with n vertices and $\bar{K}_n$ the union of n isolated vertices.

If G is a graph with $|G| = n$ and degrees $d_1 \leq d_2 \leq \dots \leq d_n$, then the sequence $(d_1, d_2, \dots, d_n)$ is called the *degree sequence* of G . A sequence of real numbers $(p_1, p_2, \dots, p_n)$ is said to be *majorized* by another sequence $(q_1, q_2, \dots, q_n)$ if $p_i \leq q_i$ for $1 \leq i \leq n$. A graph G is *degree-majorized* by a graph H if $|G| = |H|$ and the nondecreasing degree sequence of G is majorized by that of H . For example, the 5-cycle is degree-majorized by the complete bipartite graph $K_{2,3}$ since $(2, 2, 2, 2, 2)$ is majorized by $(2, 2, 2, 3, 3)$.

Ever since Dirac's theorem [4] for hamiltonicity was established, theorems for various hamiltonian properties had been derived based on the degree conditions of a given graph. Some of the well-known results are presented below.

Theorem 1. (Ore, 1960/63, [10, 11]) Let $G = (V, E)$ be a graph with $|G| = n \geq 3$. If $\deg_G(u) + \deg_G(v) \geq n + k$ for any pair of non-adjacent vertices $\{u, v\}$ in V , then G is traceable if $k = -1$, hamiltonian if $k = 0$, and hamiltonian-connected if $k = 1$.

Theorem 2. (Chvátal, 1972, [3]) Let G be a graph with $|G| = n \geq 3$. Assume that the degree d_i of G satisfy $d_1 \leq d_2 \leq \dots \leq d_n$. Then
(i) G is traceable if for every $i \leq n/2$, $d_i < i \Rightarrow d_{n+1-i} \geq n - i$ holds.
(ii) G is hamiltonian if for every $i < n/2$, $d_i \leq i \Rightarrow d_{n-i} \geq n - i$ holds.

Theorem 3. (Lick, 1970, [9]) Let G be a graph with $|G| = n \geq 3$. Assume that the degree d_i of G satisfy $d_1 \leq d_2 \leq \dots \leq d_n$. Then G is hamiltonian-connected if for every $2 \leq i \leq n/2$, $d_{i-1} \leq i \Rightarrow d_{n-i} \geq n - i + 1$ holds.

It is easy to see that, compared with degree sums, the concept of degree sequences characterizes a graph in a more refined way. Consider $G_i = (2K_2 + K_1) \vee K_{i+1}$, where $1 \leq i \leq 2$, for example. It is easy to see that $|G_i| = 6 + i$. Applying Theorem 2 (i) on G_1 and (ii) on G_2 , we know that G_1 is traceable and G_2 is hamiltonian. However, for any pair of non-adjacent vertices $\{u, v\}$ in G_i , G_1 fails to satisfy the condition $\deg_{G_1}(u) + \deg_{G_1}(v) \geq |G_1| - 1$, and G_2 fails to satisfy the condition $\deg_{G_2}(u) + \deg_{G_2}(v) \geq |G_2|$. Let G_3 be obtained by $(\bar{K}_2 + K_4) \vee K_3$ with an

additional edge between a vertex of $\bar{K}_2$ and K_4 . Then G_3 is a hamiltonian-connected graph, which satisfies Theorem 3 but fails to satisfy Theorem 1. These examples show that the degree sequence of a graph helps to determine the associated hamiltonian properties when its degree sum condition (as in Theorem 1) gives no conclusion.

Researches about the minimum number of edges that imply the desired property on graphs have attracted many attentions. See [1, 2, 5] for example, in which the minimum number of edges is given as a function of the total number of vertices of any graph. In [6] and [7], Ho and his coauthors studied the minimum number of edges for any graph to be hamiltonian or hamiltonian-connected, and expressed it as a function of $|G| = n$ and the minimum degree $\delta(G) = \delta$. Such results have many applications in interconnection networks under conditional faults, and provide better lower bounds for the number of edges by taking $\delta(G) = \delta$ into account. See [6], [7] and their references. The techniques used in [6] and [7] are based on Theorem 2 and 3, where hamiltonian properties are described in terms of the degree sequence of a given graph.

Chvátal mentioned the degree sequence behavior for a graph to remain hamiltonian after the removal of up to k faulty vertices [3]. To our knowledge, other than this study, no result about the vertex fault version of Theorem 2 and 3 has been published. Inspired by the above-mentioned works, we intend to establish the following theorem.

Theorem 4. Let G be a graph with $|G| = n \geq 3$. Assume that $(d_1, d_2, \dots, d_n)$ is the degree sequence of G . Let k and k' be integers with $0 \leq k' \leq k$, $F \subseteq V(G)$ and $|F| = k'$, and $j \leq \frac{n}{2}$.

(i) Let $0 \leq k \leq n - 2$. $G - F$ is traceable if the degree sequence $(d_1, d_2, \dots, d_n)$ satisfies

$$d_j < j + k' \leq \frac{n + k'}{2} \Rightarrow d_{n-j-k'+1} \geq n - j. \quad (1)$$

(ii) Let $0 \leq k \leq n - 3$. $G - F$ is hamiltonian if the degree sequence $(d_1, d_2, \dots, d_n)$ satisfies

$$d_j \leq j + k' < \frac{n + k'}{2} \Rightarrow d_{n-j-k'} \geq n - j. \quad (2)$$

(iii) Let $0 \leq k \leq n - 4$. $G - F$ is hamiltonian-connected if the degree sequence $(d_1, d_2, \dots, d_n)$ satisfies

$$d_j \leq j + k' + 1 \leq \frac{n + k'}{2} \Rightarrow d_{n-j-k'-1} \geq n - j. \quad (3)$$

2 Proof of the main theorem

Let n be the total number of vertices in a graph and k be an integer with $k \geq 0$. We first define three graph families as follows and give an illustration for $H_{m,n}^k$ in Figure 1.

- (a) $T_{m,n}^k = (\overline{K}_{m+1} + K_{n-2m-k-1}) \vee K_{m+k}$, where $n \geq 4$, $0 \leq k \leq n-2$, and $0 \leq m \leq \frac{n-k-2}{2}$.
- (b) $C_{m,n}^k = (\overline{K}_m + K_{n-2m-k}) \vee K_{m+k}$, where $n \geq 5$, $0 \leq k \leq n-3$, and $1 \leq m \leq \frac{n-k-1}{2}$.
- (c) $H_{m,n}^k = (\overline{K}_{m-1} + K_{n-2m-k+1}) \vee K_{m+k}$, where $n \geq 6$, $0 \leq k \leq n-4$, and $2 \leq m \leq \frac{n-k}{2}$.

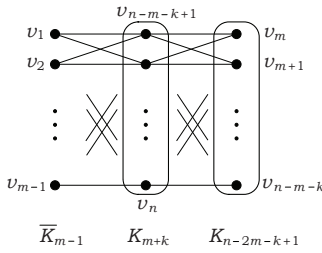


Figure 1: An illustration of $H_{m,n}^k$.

Let the faulty vertex set $F \subseteq V(K_{m+k})$ with $|F| = k$. Then it is easy to check that $T_{m,n}^k - F$ has no hamiltonian path, $C_{m,n}^k - F$ has no hamiltonian cycle, and $H_{m,n}^k - F$ has no hamiltonian path between some pair of distinct vertices $\{u, v\}$ in $K_{m+k} - F$. Thus we have the following lemma.

Lemma 1. The graph $T_{m,n}^k$ is not k -vertex fault traceable, the graph $C_{m,n}^k$ is not k -vertex fault hamiltonian, and $H_{m,n}^k$ is not a k -vertex fault hamiltonian-connected graph.

To prove Theorem 4, we recall the following theorem by Chvátal [3] first, which contains (ii) of Theorem 4. Note that any graph in the family $C_{m,n}^k$, which was introduced by Chvátal [3], has the greatest degree sequence among all graphs with the same number of vertices and being not k -vertex fault hamiltonian.

Theorem 5. (Chvátal, 1972, [3]) Let $0 \leq k \leq n-3$. If the degree sequence $(d_1, d_2, \dots, d_n)$ of a graph G satisfies

$$d_j \leq j+k < \frac{n+k}{2} \Rightarrow d_{n-j-k} \geq n-j, \quad (4)$$

then G is k -vertex fault hamiltonian. On the other hand, if the degree sequence of G fails to

satisfy (4), then it is majorized by the degree sequence of the graph $C_{m,n}^k$, which is not k -vertex fault hamiltonian.

Proof of Theorem 4: (ii) is directly derived from Theorem 5. We will show (iii) and (i) using the result of (ii).

To show (iii). Assume that the sequence $(d_1, d_2, \dots, d_n)$ satisfies (3). For any faulty vertex set $F = \{v_1, v_2, \dots, v_{k'}\}$ in G , we want to show that for any pair of vertices $\{u, v\}$ in $G - F$, there exists a hamiltonian path between u and v in $G - F$. Let G^* be a graph by in G adding a new vertex x and new lines joining x to all the vertices of $F \cup \{u, v\}$. Let $(d_1^*, d_2^*, \dots, d_{n+1}^*)$ be the degree sequence of G^* . Note that $d_1^* = \deg_{G^*}(x) = k' + 2$ since $d_2^* \geq d_1 \geq \delta(G) \geq k' + 3$, and for $1 \leq i \leq n$, $d_i \leq d_{i+1}^* \leq d_i + 1$. From (3), we have

$$\begin{aligned} d_{j-1} &\leq (j-1) + k' + 1 \leq \frac{n+k'}{2} \\ \Rightarrow d_{n-(j-1)-k'-1} &\geq n - (j-1). \end{aligned} \quad (5)$$

Note that n and k' are integers. Thus (5) is equal to

$$\begin{aligned} d_{j-1} &\leq j + k' < \frac{(n+1)+k'}{2} \\ \Rightarrow d_{(n+1)-j-k'-1} &\geq (n+1) - j. \end{aligned} \quad (6)$$

If $d_j^* = d_{j-1} + 1 > j + k'$, we don't need to check the above condition. If $d_j^* \leq j + k'$, since $d_{(n+1)-j-k'}^* \geq d_{(n+1)-j-k'-1}^*$, from (6), we have

$$\begin{aligned} d_j^* &\leq j + k' < \frac{(n+1)+k'}{2} \\ \Rightarrow d_{(n+1)-j-k'}^* &\geq (n+1) - j. \end{aligned} \quad (7)$$

Therefore, according to (7), the degree sequence $(d_1^*, d_2^*, \dots, d_{n+1}^*)$, with $n+1$ in place of n , satisfies (2). By (ii), $G^* - F$ is hamiltonian. Since that $G^* - F$ has a hamiltonian cycle if and only if $G - F$ has a hamiltonian path between u and v , $G - F$ is hamiltonian-connected.

The derivation for (i) is left for readers and is skipped here. The proof is complete. $\square$

According to the three results of Theorem 4, we have the following equivalent theorem.

Theorem 6. Let G be a graph with $|G| = n \geq 4$ and $k \geq 0$ be an integer. Assume that $(d_1, d_2, \dots, d_n)$ is the degree sequence of G .

(i) If G is not k -vertex fault traceable, where $n \geq 4$ and $0 \leq k \leq n-2$, then there exists at least one

integer $0 \leq m \leq \frac{n-k-2}{2}$ such that $d_{m+1} \leq m+k \leq \frac{n+k-2}{2}$ and $d_{n-m-k} \leq n-m-2$.

(ii) If G is not k -vertex fault hamiltonian, where $n \geq 5$ and $0 \leq k \leq n-3$, then there exists at least one integer $1 \leq m \leq \frac{n-k-1}{2}$ such that $d_m \leq m+k \leq \frac{n+k-1}{2}$ and $d_{n-m-k} \leq n-m-1$.

(iii) If G is not k -vertex fault hamiltonian-connected, where $n \geq 6$ and $0 \leq k \leq n-4$, then there exists at least one integer $2 \leq m \leq \frac{n-k}{2}$ such that $d_{m-1} \leq m+k \leq \frac{n+k}{2}$ and $d_{n-m-k} \leq n-m$.

It can be shown that the graph families $T_{m,n}^k$, $C_{m,n}^k$, and $H_{m,n}^k$ have the greatest degree sequences among all graphs with the same number of vertices and being not k -vertex fault traceable, not k -vertex fault hamiltonian, and not k -vertex fault hamiltonian-connected, respectively. It leads to the following corollary.

Corollary 1. Adding an extra edge to $T_{m,n}^k$, $C_{m,n}^k$, and $H_{m,n}^k$ results in a k -vertex fault traceable graph, a k -vertex fault hamiltonian graph, and a k -vertex fault hamiltonian-connected graph, respectively. Consequently, if G is not k -vertex fault traceable, then G is degree-majorized by $T_{m,n}^k$; if G is not k -vertex fault hamiltonian, then G is degree-majorized by $C_{m,n}^k$; if G is not k -vertex fault hamiltonian-connected, then G is degree-majorized by $H_{m,n}^k$.

3 Conclusion

In this paper, we establish a theorem, Theorem 4, based on graphic degree sequences. In Theorem 4, the sufficient conditions for any graph G to be k -vertex fault traceable, k -vertex fault hamiltonian, and k -vertex fault hamiltonian-connected are given, respectively. We use this theorem to write down the greatest degree sequence for any graph being not k -vertex fault traceable (or not k -vertex fault hamiltonian, not k -vertex fault hamiltonian-connected, resp.) and discover the corresponding graph family $T_{m,n}^k$ (or $C_{m,n}^k$, $H_{m,n}^k$, resp.). Theorem 4 takes faulty vertices into consideration and improves the known results, so Theorem 2 and 3 become special cases for Theorem 4 with $k = 0$. Currently, we are exploring the possibility of extending our derivation to obtain parallel results for edge-fault tolerant hamiltonian properties. Namely, we intend to establish degree sequence characterization for hamiltonian properties under edge faults. We hope a stronger result

for mixed-fault tolerant hamiltonicity can be established in the near future.

References

- [1] B. Bollobás and A. Thomason, “Weakly pancyclic graphs”, *J. Combinatorial Theory Series B*, Vol. 77, pp. 121–137, 1999.
- [2] S. Brandt, “A sufficient condition for all short cycles”, *Discrete Applied Mathematics*, Vol. 79, pp. 63–66, 1997.
- [3] V. Chvátal, “On Hamilton’s ideal”, *J. Combinatorial Theory Series B*, Vol. 12, pp. 163–168, 1972.
- [4] G. A. Dirac, “Some theorems on abstract graphs”, *Proceedings of the London Mathematical Society*, Vol. 3, pp. 69–81, 1952.
- [5] P. Erdős, T. J. Reid, R. Schelp, and W. Staton, “Sizes of graphs with induced subgraphs of large maximum degree”, *Discrete Mathematics*, Vol. 158, pp. 283–286, 1996.
- [6] T. -Y. Ho, C. -K. Lin, Jimmy J. -M. Tan, D. Frank Hsu, and L.H. Hsu, “On the extremal number of edges in hamiltonian-connected graphs”, *Applied Mathematics Letters*, Vol. 23, pp. 26–29, 2010.
- [7] T. -Y. Ho, C. -K. Lin, Jimmy J. -M. Tan, D. Frank Hsu, and L.H. Hsu, “On the extremal number of edges in hamiltonian graphs”, *J. Information Science and Engineering*, Vol. 27, pp. 1659–1665, 2011.
- [8] S.-S. Kao, K.-M. Hsu, and L.-H. Hsu, “Cubic planar Hamiltonian graphs of various types”, *Discrete Mathematics*, Vol. 309, pp. 1364–1389, 2006.
- [9] D. R. Lick, “A sufficient condition for hamiltonian connectedness”, *J. Combinatorial Theory*, Vol. 8, pp. 444–445, 1970.
- [10] O. Ore, “Note on Hamilton Circuit”, *American Mathematical Monthly*, Vol. 67, pp. 55, 1960.
- [11] O. Ore, “Hamilton-Connected Graphs”, *Journal de Mathématiques Pures et Appliquées*, Vol. 42, pp. 21–27, 1963.